

TraceLens AI

Autonomous Microservice Incident Triage & Root-Cause Localization Engine

Category: AIOps & Cloud Reliability

Tech Stack: GNNs + DAG Critical Path + LLM Agents

Live Demo: Streamlit Cloud / GitHub

Target MTTR Reduction: 85%

1. The Problem: The High Cost of Alert Storms

Modern microservices cause catastrophic alert noise and delayed triage

\$9,000 / Minute Average cost of enterprise downtime for Tier-1 services (Gartner).	70% MTTR Waste SREs spend over 45 minutes manually correlating spans and logs across siloed tools.
Alert Storm Blindness A single database deadlock fires 50+ secondary alerts across downstream services.	Legacy Tool Gaps Datadog and Dynatrace alert on symptoms, not the mathematical root-cause culprit.

2. The Solution: Hybrid Graph AI & SRE Agent

Deterministic graph mathematics combined with LLM incident reasoning

Step 1: DAG Ingestion	Parses distributed OpenTelemetry & Jaeger traces into Directed Acyclic Graphs.
Step 2: Critical Path Slack	Calculates earliest/latest schedules to filter out non-blocking asynchronous noise.
Step 3: Edge-GAT Neural Net	Propagates inter-service network latency and error status to rank root-cause culprit spans.
Step 4: Agentic Remediation	Synthesizes graph telemetry with logs to auto-generate exact kubectl / SQL fix scripts.

3. Technical Originality: Beyond Prompt Wrappers

Why TraceLens achieves zero-hallucination, predictable diagnosis

Dimension	Generic LLM Wrappers	TraceLens AI (Ours)
Graph Topology	Ignored (flattened to raw text)	Preserved as DAG with edge latency weights
Context Window	Overflows on large 1,000+ span traces	Pre-filtered algorithmically via Critical Path
Output Precision	Prone to hallucinated microservices	Mathematical proof of bottleneck culprit
Actionability	Generic text advice ('investigate DB')	Executable bash / kubectl recovery commands

4. Market Opportunity & Business Model

High-margin B2B SaaS in a rapidly expanding observability landscape

Market Sizing (AIOps & Observability) • TAM: \$61.3B Cloud Observability by 2028 (11.7% CAGR) • SAM: \$14.2B Cloud-Native & Kubernetes Telemetry • SOM: \$1.8B Mid-to-Enterprise DevOps Teams	Tiered Revenue Model • Developer Tier (Free / OSS): Local CLI & UI up to 10k spans/mo. • Team Tier (\$49/seat/mo): Real-time Slack/PagerDuty bot. • Enterprise (\$0.05/1k spans): On-prem GNN inference & automated Kubernetes remediation.
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5. Competitive Advantage Matrix

How TraceLens outclasses legacy observability vendors

Capability	Datadog / Dynatrace	PagerDuty / incident.io	TraceLens AI
Root-Cause Engine	Thresholds & Heuristics	Manual SRE Triage	Mathematical DAG GNN
Remediation Scripts	No	Runbooks (Manual)	Automated Kubectl/Bash
Pricing Model	High Host/GB Tax	Per User Seat	Usage-based Spans
Mean Time to Triage	15 - 45 min	20 - 40 min	< 5 Seconds

6. Roadmap & Execution Milestones

From intelligent assistant to autonomous closed-loop self-healing

Phase 1: Now (Hackathon MVP)	• Streamlit Cloud Web UI + Chaos Simulator • OpenTelemetry / Jaeger DAG Ingestion • Groq LLM SRE Report & Remediation Generator
Phase 2: Q3 2025 (Integrations)	• Bidirectional Slack & PagerDuty bot integrations • GitHub Actions CI/CD trace regression detection
Phase 3: Q4 2025 (Closed-Loop)	• Kubernetes Self-Healing Operator (auto-restart, circuit breaking) • eBPF kernel-level zero-code instrumentation

Transforming Cloud Observability with TraceLens AI

Reducing MTTR from 45 minutes to 5 seconds with Graph AI and Automated Remediation.

[Live Demo: Accessible on Streamlit Cloud](#)

[GitHub Codebase: Fully open-source with unit tests & GNN engine](#)

Thank you! Questions & Discussion